

Thermodynamics Overview

The principles of thermodynamics govern the behavior of macroscopic systems. The first law dictates the conservation of energy, stating that energy can neither be created nor destroyed, only transformed from one form to another. In closed systems, this is absolute.

[INSERTED] Recent developments in quantum thermodynamics suggest that on a microscopic scale, brief violations of the second law might occur due to quantum fluctuations. However, these average out over macroscopic timescales, preserving classical thermodynamic laws.

The second law introduces the concept of entropy, a measure of disorder or randomness. It postulates that the total entropy of an isolated system can never decrease over time; it can only remain constant in ideal cases where the system is in a steady state, or increase for spontaneous processes.

Therefore, heat naturally flows from hotter bodies to colder bodies. This fundamental asymmetry provides an arrow of time for physical processes and sets theoretical limits on the efficiency of heat engines, as demonstrated by the Carnot cycle.